

Subject Code	AF5351
Subject Title	Derivatives Securities
Credit Value	3
Level	5
Normal Duration	One Semester
Pre-requisite / Co-requisite/ Exclusion	Pre-requisite: Investments (AF5344)
Objectives	This subject contributes to the achievement of the Master of Finance Programme Outcomes by enabling students to solve asset management/corporate finance problems (Outcome 2).
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - have an in-depth understanding of the derivative assets such as options, futures, and forwards; - price and formulate different trading strategies of derivatives traded in the financial market; - use derivative assets in hedging and trading from the perspectives of a corporate treasurer or trader; - construct and price complex derivative financial instruments; and - use technology related to data science to construct, price and analyze trading strategies involving derivative securities
Subject Synopsis/ Indicative Syllabus	Derivative Assets and Markets Characteristics of forward, futures, and options; market structures and conventions. Pricing and Trading Strategies of Futures Properties of forward and futures prices; forward and futures pricing model; futures trading strategies. Pricing and Trading Strategies of Options The Binomial model; the Black-Scholes Model: assumptions, adjustments and applications; option trading strategies including spreads, straddles, straps and strips. Hedging and Trading Strategies for Options and Futures Hedging concepts; types of hedges; determination of hedge ratios.
Teaching/Learning Methodology	Most of the material will be covered in a lecture format but class participation is strongly recommended for students to obtain the most out of this course.

Assessment Methods in Alignment with Intended Learning Outcomes

To assess whether the students achieved the learning outcomes of this subject, the focus of mid-term examination will be on the use and the principle of pricing of forward and futures. Students are also required to do a group/individual assignments to make sure that they understand each week's materials. The final examination will cover all the materials before and after the midterm, including the pricing and formulation of the trading strategies of derivative instruments and the usage of derivative securities in the hedging and trading from a corporate treasurer or trader's perspective.

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
		a	b	c	d	e	
1. Group Project	20%	✓	✓	✓	✓	✓	
2. Midterm	30%	✓	✓	✓	✓		
3. Attendance & Participation	10%	✓	✓	✓	✓		
4. Final Examination	40%	✓	✓	✓	✓		
Total	100 %						

Note: The specific requirements on each assessment component discussed above could be adjusted based on the pedagogical needs of subject lecturers.

To reflect the significant technology content in this subject, 10% of the overall weighting of this subject is based on individual assessment concerning technology-related knowledge.

Student Study Effort Expected	Class contact:	
	▪ Seminars	39 Hrs.
	Other student study effort:	
	▪ On average, students are expected to spend around 4 hours for reading materials/ textbook and to answer questions and solve numerical problems a weekly basis.	52 Hrs.
	▪ Preparations for exams	26 Hrs.
	Total student study effort	117 Hrs.
Reading List and References	<u>Indicative Reading</u> Hull, J., <i>Options, Futures, and Other Derivatives</i>, Global 11th edition, Pearson	